A Mapping Approach for *Eucalyptus* Plantations Canopy and Single Tree Using High-Resolution Satellite Images in Liuzhou, China

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***Abstract*— Accurate canopy and single-tree mapping is important to obtain information on the ecological structure and biogeophysical parameters for forests. Although some airborne radars can retrieve canopy and single-tree information within a smaller area, the optical satellite imagery-based approaches for rapidly and accurately mapping them over a large region are still limited. In this study, based on *Eucalyptus* canopy and single-tree texture and spectral features, we proposed a mapping approach using the combinations of image morphology, the Otsu method, and an adaptive iterative erosion algorithm (EUMAP). Then, we applied the commonly used red/green/blue bands from the high-resolution satellite images, which are freely available, to map the canopy and single tree in *Eucalyptus* plantations in southern China. EUMAP consists of two steps: 1) *Eucalyptus* canopy identification for various canopy density regions and 2) adaptive iterative erosion to separate single tree. Our study was conducted in Chengzhong and Liubei, Liuzhou, China. The accuracy evaluation was carried out in the state-owned Sanmenjiang forest farm. The results showed that the average *F*1 score for mapping canopy and single tree reached 88.34% and 86.40%, respectively. For the whole study area, there were 7 033 021 *Eucalyptus* trees and the average density was 819 trees per hectare. The approach adopted in this study, combining the prior knowledge about image morphology and single-tree texture features of *Eucalyptus* plantations, was highly efficient for satellite image processing and had excellent applicability to large-scale *Eucalyptus* plantation mapping. Our study highlights the necessity of prior knowledge for forest mapping using satellite images without requiring a training sample and provides a universal approach of accurate large-scale mapping for specific forest species with common red/green/blue images.**

***Index Terms*— Adaptive iterative erosion, *Eucalyptus* plantations, high-resolution satellite image, image morphology, single-tree mapping.**

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I. Introduction

EUCALYPTUS plantations are exploitable forests representative of southern China, with a significant economic and ecological impact. They contribute strongly to forest conservation, climate regulation, and carbon mitigation (both in terms of peak date and net emissions) [1], [2]. In addition, *Eucalyptus* wood processing has become an important and profitable industry in some areas of China [3], [4]. However, a recent marked increase in timber demand in China has caused unusually rapid development of *Eucalyptus* plantations [5], [6]. Concurrently, serious related problems have emerged, including the decline of soil fertility and biodiversity [7]. The lack of data on *Eucalyptus* plantations affects the ability of the Chinese government to enact informed management decisions [8]. Accurate and rapid acquisition of the specific quantity and location data of single tree in *Eucalyptus* plantations can help governments and forest managers effectively carry out forest resource management, ecological protection, and timber economic planning [9], [10]. In addition, regular monitoring of the canopy and single tree can help plantation forestry managers and individual growers understand the growth rate, health status, and expected yield of *Eucalyptus* plantations while helping to promote the scientific development of sustainable forestry and protection [11], [12]. Therefore, fast collection of accurate information on *Eucalyptus* plantations is important to assess the balance between their adverse ecological impact and positive socioeconomic value.

Since Crowther et al. [13] discussed the spatial distributions of trees at the global scale, this issue has been extensively studied, and the study data are basically obtained from high-resolution remotely sensed images from drones or satellites [14], [15], [16], [17]. Unfortunately, the most accurate images, deriving from drone data, are limited in scale, whereas satellite images could provide information at the global scale [18], [19], [20]. In addition, an accurate and computationally efficient approach is essential for

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large-scale canopy and single-tree mapping. An issue in whole-canopy and single-tree mapping is that canopy density depends on tree age [21]. Differences in artificial planting techniques and seedling development also influence canopy density in *Eucalyptus* plantations [22], [23]. Therefore, it is crucial to construct an accurate and computationally efficient approach for mapping canopy and single tree at a large scale. Current common mapping methods are: traditional image processing algorithms, machine learning, and deep-learning models. Traditional image processing algorithms usually include basic methods, such as local maximum filtering, image segmentation, image binarization, and template matching [24], [25]. Some studies have proved that combining two or more methods may perform better results to achieve detecting and delineating tree canopy [26], [27]. The earlier machine-learning techniques include classifier training and feature extraction. Features can be extracted manually using techniques such as the histogram of oriented gradient and scale-invariant feature transform [28], [29]. Or, the target features can be generated using methods such as vegetation spectral indices and texture characteristics [30], [31]. The decision tree, random forest, and multilayer perceptron are widely used too [32], [33]. Deep-learning methods have recently been applied to many fields of study [34], [35]. Wang et al. [36] investigated deep learning for rubber-tree canopy mapping. They converted data from lidar into images by orthogonal transformation and then used the images as input for a deep-learning network to achieve accurate mapping of a rubber-tree canopy. Weinstein et al. [37] proposed a semisupervised method to generate annotated data using unsupervised algorithms and point-cloud data to alleviate the difficulty of data annotation in single-tree canopy mapping. Weinstein et al. [38] also investigated the generalization capabilities of deep-learning methods for canopy detection in different types of forests.

However, to achieve a certain precision, both machine- and deep-learning methods necessitate a significant amount of sample selection, learning, model training, and classification [39], [40]. Because the computational requirements for data processing are extremely intensive, the applications are limited at a large scale. These restrictions make it rare for current studies to map more than million-level trees [41]. There are also some mapping problems in the study of mapping million-level trees. Including mapping efficiency and high canopy density forest areas can only be mapped as closed tree canopies, and single trees cannot be separated [42]. In addition, the neural network-based deep-learning methods are similar to a black-box model, whose interpretation is poor and hard to provide a clear generalization for a particular task [43]. In contrast, when the features of the extracted object can be precisely designed with the prior knowledge of human cognitive aspects of things, it can show advantages of rapidity and high efficiency when combined with well-established and widely recognized traditional image processing methods.

In this study, to enable faster and more efficient mapping trees from remote sensing images at a large scale, we improved the traditional image extraction method based on image morphology and proposed an approach for mapping *Eucalyptus* canopy and single tree (EUMAP). The EUMAP uses an adaptive iterative erosion approach to avoid interference from canopies overlap caused by image processing, which are really important in the remote sensing single-tree detection domain.

II. Materials and Methods

A. Study Area and Data Description

Liuzhou is located in the south of China, which belongs to the central subtropical climate zone, with an annual average temperature of 20 °C, an annual precipitation of 1300–1700 mm, and an annual evaporation of 1471–1750 mm. The benefits from abundant rainfall and warm temperatures provide a favorable environment for plant growth [44]. Our study area is Chengzhong and Liubei, Liuzhou, with a total area of 380 km^2, including 72 km^2 of *Eucalyptus* plantations based on government survey. The *Eucalyptus* plantations used in this study were provided by the state-owned forest farms, which covers various areas from the inner city to city border and even the hills outside the city (see Fig. 1).

The high-resolution remote sensing satellite images used in this study were acquired by the WorldView-3 satellite. WorldView-3, launched on 13 August 2014, is the world's first commercial optical remote sensing satellite. The orbital height is 617 km, the swath width is 13.2 km, the revisit period is one day, and the resolution of panchromatic and multispectral images are approximately 0.3 and 1.2 m, respectively. Sixty 0.3-m resolution satellite images were acquired in the conventional red, green, and blue (RGB) spectral bands on 6 May 2020 and 24 July 2021 in the study area. Each image covered an area of 6.9 km^2. The width of each image was 3.24 km^2 and the height was 2.14 km^2, resulting in an image capacity of 1.04 GB. The total data size for this study was 62.4 GB.

B. Field Survey

We conducted a field survey of the study area in July 2022. In the study area, 12 *Eucalyptus* plantation areas were selected as survey sample areas [see Fig. 1(b)]. Within these delineated plots, a comprehensive survey was carried out, spanning a total of 56 single *Eucalyptus* trees (see Fig. 2). These trees exhibit diverse distribution patterns and a spectrum of ages, enhancing the representativeness of the collected data (see Table I). The scope of this survey encompassed a comprehensive evaluation of both the physical attributes of the trees and their surrounding living environments. Among them, the longitude, latitude, and altitude of the *Eucalyptus* were recorded using a GPS locator (GARMIN GPSMAP-679). The diameter at breast height, tree height, and canopy width of *Eucalyptus* was measured with tape measure and latitude and longitude laser compass (DQL-16ZJ).

C. Overview

The approach of mapping *Eucalyptus* canopy and single tree (EUMAP) used in this study is based on image morphology [45], which is usually used to analyze spatial

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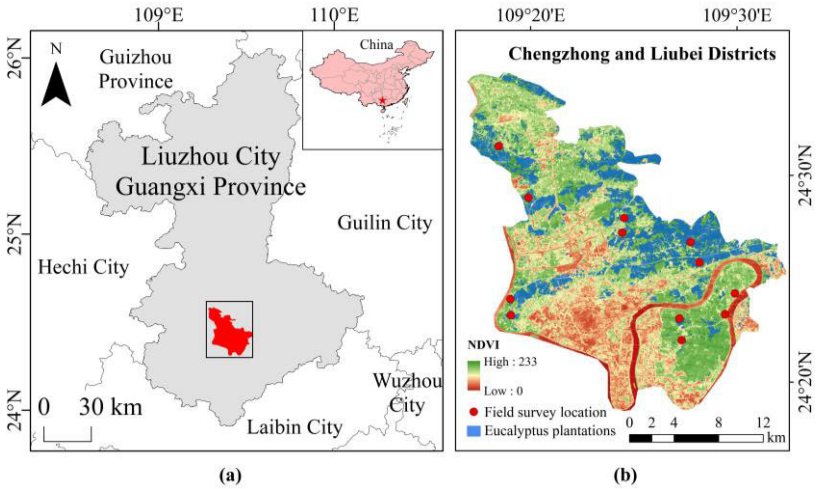


Fig. 1. Study area and *Eucalyptus* plantations. (a) Geographical location of the Chengzhong and Liubei, Guangxi, China (red region). (b) Normalized difference vegetation index (NDVI), field survey location, and *Eucalyptus* plantations (blue region) in the study area.



Fig. 2. Photographs of *Eucalyptus* plantations in the field with *Eucalyptus* physical parameters and drone survey work.

TABLE I
PHYSICAL PARAMETER DATA OF *Eucalyptus* FIELD SURVEY

	Tree Height (m)	Canopy Width (m)	DBH (cm)
Max	18.75	6.84	18.21
Min	5.31	1.14	5.28
Average value	12.95	3.29	10.99

structure based on geometric theory, integral geometry, and mesh algebra. The key principle of EUMAP is to integrate the image features of various canopy density regions of *Eucalyptus* plantations with the grayscale feature performance between canopies and shadows, so as to perform adaptive iterative erosion by providing the connection of different

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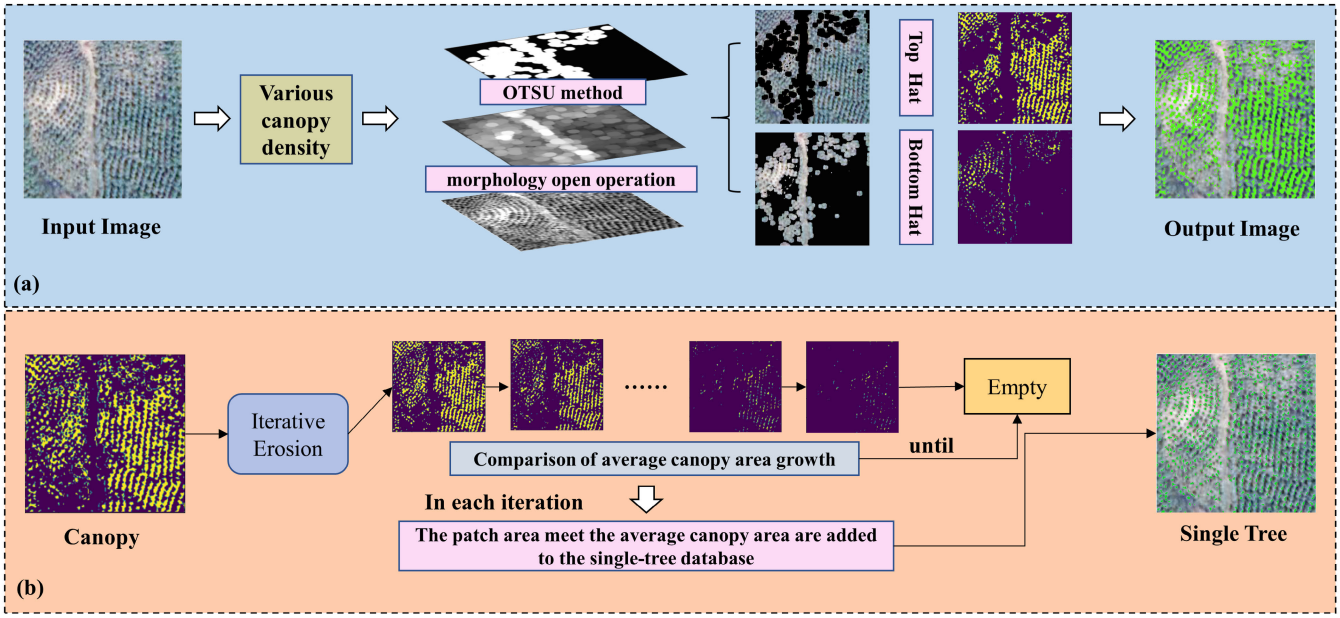


Fig. 3. Framework of the proposed *Eucalyptus* mapping approach (EUMAP), including two steps. (a) Canopy identification in various canopy density regions. (b) Adaptive iterative erosion to separate single tree.

feature processing methods. EUMAP can be generally divided into two steps: 1) canopy identification for various canopy density regions and 2) adaptive iterative erosion to separate single tree (see Fig. 3). This is an unsupervised approach, and thus, it does not need a lot of sample data and training. In the field of tree canopy and single-tree extraction, EUMAP can be used to solve the bottleneck of detecting small and dense tree crowns in remote sensing images that traditional image processing methods struggle with. Furthermore, it also has the advantage of efficiently processing large-scale data compared to machine learning and deep learning.

D. Canopy Identification in Various Canopy Density Regions

EUMAP is used to identify canopy in various density regions mainly based on the dilation and erosion of image morphology. First, the RGB color index system is based on the colors recognized by the human eye. However, since we mainly focus on the light and shade changes between canopy and shadows, the RGB color index system cannot express information intuitively and accurately. In this study, the lightness (L) index of the hue/saturation/lightness color space was extracted from the RGB images. The equation for color transformation is [46]

$$L = 0.5[\max(R, G, B) + \min(R, G, B)] \tag{1}$$

where R, G, and B are the grayscale values of RGB images. Then, we define an input image $f(s, t)$, a structuring element $b(x, y)$, and their definition domains D_f and D_b. The grayscale dilation operation, noted $f \oplus b$, is defined as

$$(f \oplus b)(s, t) = \max\left[f(s\text{-}x, t\text{-}y) + \frac{b(x, y)}{(s\text{-}x, t\text{-}y)}\right] \in D_f. \tag{2}$$

Grayscale dilation enlarges the bright areas of the input image to enhance their edges. At the same time, dark pixels contained in the structuring elements are eliminated and the number of remaining dark pixels also decreases.

The grayscale erosion operation, noted $f \ominus b$, is defined as

$$(f \ominus b)(s, t) = \min\left[f(s\text{-}x, t\text{-}y) - \frac{b(x, y)}{(s\text{-}x, t\text{-}y)}\right] \in D_f. \tag{3}$$

Grayscale erosion is the opposite of dilation, darkening the output image relative to the input image. If bright areas in the input image are smaller than the structuring element, image brightness is reduced.

Dilation and erosion in morphological image processing can be combined into the opening operations [47]. The opening operation consists in eroding image features and then dilating the eroded features. It is used to eliminate small objects, separate objects in narrow areas, and smooth the boundaries of larger objects without excessively modifying their area. Also, the *Eucalyptus* plantations with various canopy densities showed significant differences in regional grayscale on satellite images. To accurately extract the *Eucalyptus* plantations with various canopy densities, it is necessary to separate the canopy density. The image opening operation was used to remove background noise: 1) shadows merged with canopies in the high canopy density area and 2) shadows merged with ground in the low canopy density area. After the opening operation to remove the noise, we used Otsu's method to effectively separate canopy regions with various canopy densities [see Fig. 3(a)]. Otsu method also known as the maximum inter-class variance method is an adaptive threshold determination method first proposed by Otsu in 1979, which derives from the least-squares principle [48], [49]. It categorizes all image pixels into two classes: background pixels and foreground pixels [50]. Applying statistical probabilities, pixel variance gives a measure of the grayscale distribution uniformity: larger variance indicates larger differences between distinct image

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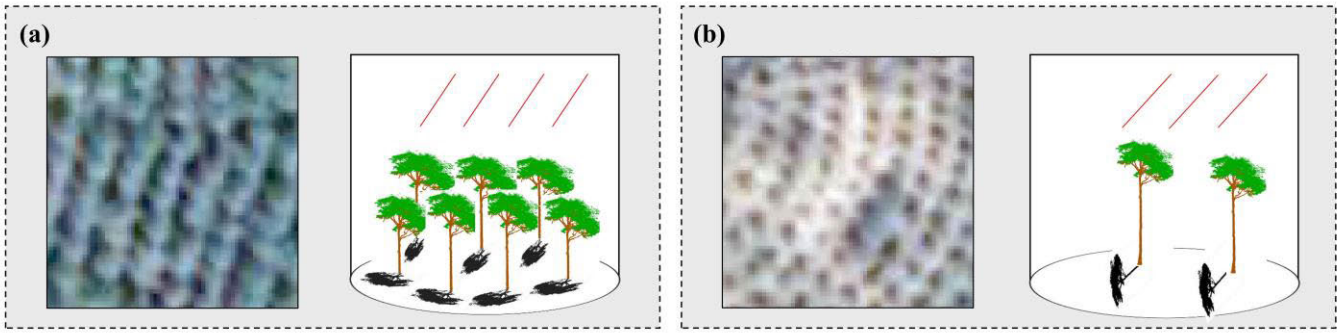


Fig. 4. *Eucalyptus* plantations with various canopy densities show different grayscale values of the canopy and the intercanopy parts. (a) Image with high canopy density. (b) Image with low canopy density.

regions. Therefore, if the variance between the two pixel classes is maximized, the misclassification error in the image is minimized [51].

For the separated *Eucalyptus* plantations with various canopy densities, this shows different grayscale values. In the areas with high canopy density, the *Eucalyptus* are closely connected, and the canopies are close to each other and even stack together, resulting in low-value shadows between canopies and bright high-value canopies on satellite images [see Fig. 4(a)]. Conversely, in the areas with low canopy density, *Eucalyptus* is scattered and independent, resulting in low-value canopies and bright high-value bare land on satellite images [see Fig. 4(b)]. *Eucalyptus* canopy exhibits a texture feature similar to palmprint-like or fingerprint-like. Based on this special texture feature and the differences in the performance of *Eucalyptus* plantations at various canopy densities, we further used the top and bottom-hat transform operations. Also, the compound opening and closing operation are used to further define the top- and bottom-hat transforms [52]. In areas with high canopy closure, the top-hat transform was used to identify high-value areas in the image. In areas with low canopy closure, the bottom-hat transform was used to identify small "holes" in the image, representing single-tree canopies in a clearing. Subsequently, results for both classes were combined to derive the complete canopy map.

E. Adaptive Iterative Erosion to Separate Single Tree

For the identified *Eucalyptus* canopies with various canopy densities, we used an adaptive iterative erosion approach to separate connected and overlapping canopies into single tree [see Fig. 3(b)]. The selection and use of parameters in the erosion algorithm are important to ensure high mapping accuracy. According to the canopy width data of *Eucalyptus* in field surveys and *Eucalyptus* plantations, which are characterized by a round or oval single-tree canopy shape, it is estimated that a single-tree canopy area is about 4–10 m^2. In addition, trees are often planted in dense rows, causing canopies to overlap or connect. Using the erosion algorithm, canopy boundaries contracted. Therefore, canopy areas notably different from the average canopy area for *Eucalyptus* were divided into two or more single-tree canopies, whereas canopies with limited contact were correctly separated. Then, the single-tree canopy

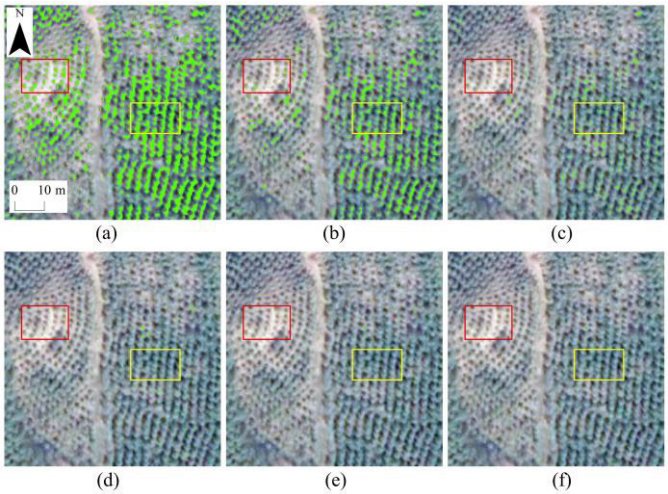


Fig. 5. Erosion results are shown for kernel values of (a) 2, (b) 4, (c) 6, (d) 8, (e) 10, and (f) 12.

vectors determined after removal of fragmented canopies were converted to point positions to identify single-tree. However, the kernel parameter values selected for erosion processing had a marked impact on mapping accuracy. Kernel parameter values constrain the calculation range for each pixel value during erosion processing. Small kernel values prevent excessive erosion of the canopy, but they cannot effectively eliminate canopy overlap. Conversely, large kernel values eliminate overlap between canopies, but smaller canopies cannot be distinguished from the background; thus, they are completely removed during erosion processing. Small kernel values, therefore, result in invalid canopy separation, whereas large kernel values cause over separation of the single-tree canopies. When the kernel value was set to 2, canopy erosion was not sufficient and some overlap remained [yellow box, Fig. 5(a)]. This problem was solved for kernel values of 4 or larger [yellow boxes, Fig. 5(b)–(f)], but excessive erosion then occurred in the bright areas [red boxes, Fig. 5(b)–(f)]. This implies that there is no optimal kernel value to correctly derive single-tree positions from canopy erosion in *Eucalyptus* plantations with different canopy sizes and closures.

Furthermore, using a constant kernel value of 2 for iterative erosion canopy processing (see Fig. 6) allowed for better control of the erosion intensity and for more gradual canopy reduction, but not for accurate control of the erosion status

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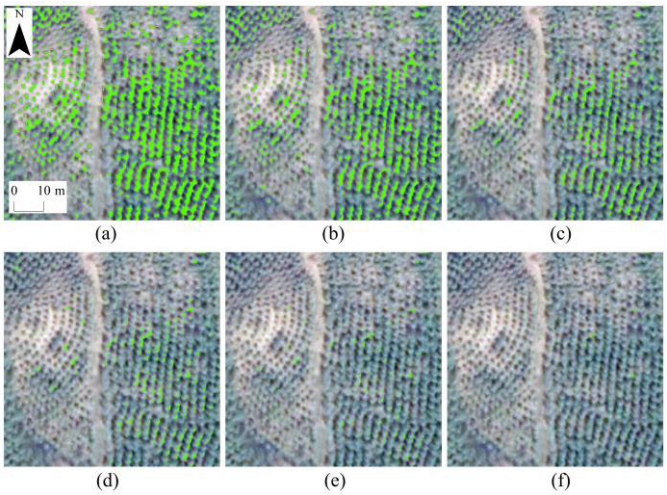


Fig. 6. Iterative erosion results for a kernel value of 2 applied to the full image at each iteration. (a)–(f) Intermediate results after iterations 1–6, respectively.

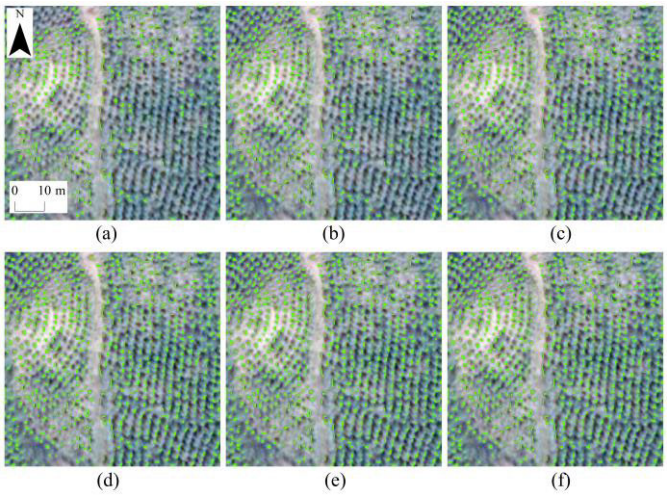


Fig. 7. Iterative erosion results for a kernel value of 2 with EUMAP, using a different number of iterations for distinct image regions. (a)–(f) Intermediate results after iterations 1–6, respectively.

of single-tree canopies. This caused smaller canopies to erode continuously, simultaneously with larger canopies, even after the expected erosion had been achieved.

To address the differences in the erosion process of canopies of different sizes and degrees of overlap, an adaptive iterative erosion approach for single-tree mapping was implemented in this study. In this approach, the kernel parameter was set to 2, but the number of iterations for canopy erosion was adaptive, which depended on the processed image itself. Adaptive erosion is given as follows: the first erosion was performed from the canopy identified by EUMAP. After each erosion iteration, the patch area that meets the average *Eucalyptus* canopy area was added to the single-tree database. Canopy patches larger than 10 m^2 are not directly rejected but are eroded and gradually smaller over the course of iteration until canopy area requirements are determined to meet canopy area requirements. In addition, considering that some fine *Eucalyptus* and normal *Eucalyptus* trees reduce their area during erosion, the lower limit of canopy area requirement extends to 0.5 m^2 to identify canopy patches smaller than 4 m^2. This enables the smaller canopies to independently separate from iterations when they meet the area requirement for the first time, while the larger canopies need to the process of continued iterative erosion due to the overlap between canopies. As the number of iterations increases, the amount of single-tree results obtained also increases. For different images, different numbers of iterations were generated adaptively during the iterative erosion process. Until all canopies in the image meet the area requirements, canopies selected at each iteration were combined to produce the final mapping result (see Fig. 7). Also, the number of iterations at the end time is the adaptive iteration number of the area. The combination of different numbers of erosion iterations yielded accurate identification of different-size canopies that were connected or overlapping while limiting the removal of smaller canopies by excessive erosion.

F. Accuracy Evaluation

We select the state-owned Sanmenjiang forest farm in the study area as the test area to verify the performance

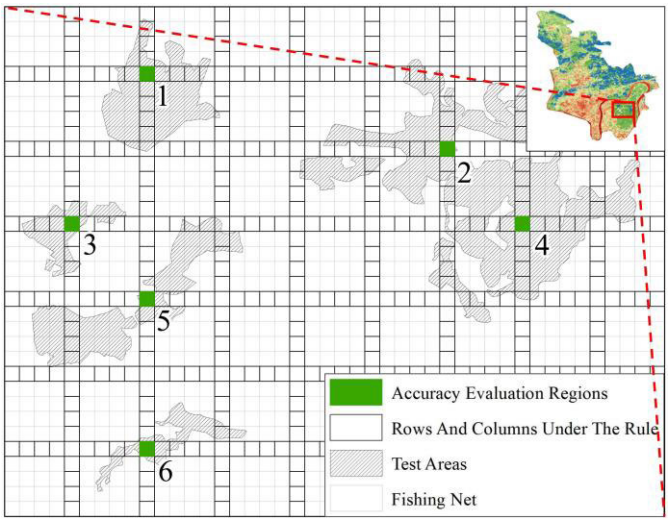


Fig. 8. Regular gridding and selection of accuracy evaluation regions in the test area.

of EUMAP. To evaluate the accuracy of EUMAP, we divided the test area into a regular grid with square mesh elements of 100 × 100 m. The row and column rules were set every five elements. Six accuracy evaluation regions were selected at rule intersections within the test area (see Fig. 8). We visually interpreted the number and location of *Eucalyptus* in the six accuracy evaluation regions as the actual reference results. In addition, we compared both the accuracies and efficiencies between our method and the state-of-the-art (SOTA) method proposed by Prof. Martin Brandt's team, which has been applied to tree mapping in Africa and Europe [17], [53], [54], [55], [56].

Here, we used the recall, precision, and $F1$ score metrics in the accuracy evaluation areas for evaluation [57], [58]. Recall (R) represents the ratio of the *Eucalyptus* correctly mapped by the algorithm to the ground truth *Eucalyptus* in the experimental region. Recall values closer to 1 indicate better mapping results. Precision (P) represents the ratio of the *Eucalyptus* correctly mapped by the algorithm to the total *Eucalyptus* mapped by the algorithm. Values closer to 1 indicate a more precise algorithm. The $F1$ score evaluates

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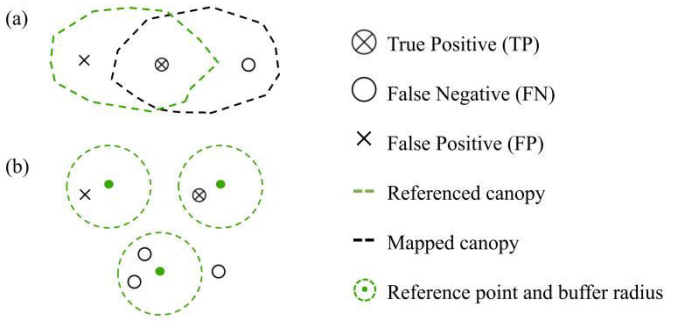


Fig. 9. Schematic description of the accuracy evaluation for canopy and single tree mapping. (a) TP, FN, FP status of canopy. (b) TP, FN, FP status of single tree. The referenced canopy represents the manually delineated canopy and the reference point represents the manually delineated position of single tree.

the global mapping quality. Values closer to 1 indicate better overall algorithm performance. The aforementioned metric can be formulated from true positives (TP), false positives (FP), and false negatives (FN) as

$$R = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN}) \tag{4}$$

$$P = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP}) \tag{5}$$

$$\text{F1 score} = 2\mathrm{R} \times P/(P + R). \tag{6}$$

For canopy identification, TP is the correctly mapped canopy area, FN is the canopy area mapped as noncanopy, and FP is the noncanopy area mapped as canopy [see Fig. 9(a)]. For single-tree identification, TP represents the number of correct mapping points, for which a unique mapped point is contained within a buffer radius of a reference point location. FN represents the number of missed detections, for which there is no mapping point within the buffer radius of a reference point location. FP means that the buffer radius of the reference point contains two or more mapping points and the points outside the buffer radius [see Fig. 9(b)].

The buffer radius in the accuracy evaluation will affect the $F1$ score. As the buffer radius increases, the number of mapped points falling into the buffer area of the reference point increases, causing TP to become larger, FN to become smaller, and the $F1$ score to become larger. On the other hand, it also increases the number of two or more mapped points appearing in the same buffer, which will cause FP to become larger and the $F1$ score to become smaller. Thus, only when the buffer radius is at a suitable size, it has the minimal overall impact on the $F1$ score. In order to accurately find the appropriate buffer radius, we started with a buffer radius of 0.5 m and gradually increased it to 3.5 m and analyzed the influence of the buffer radius of the reference point on the three indices (precision, recall, and $F1$ score) for the test area. Table II lists the three indices' results corresponding to different buffer radii. Through testing, we found that the average $F1$ score reaches a peak when the buffer radius is 2.5 m. Thus, we chose 2.5 m as the final buffer radius for our experiment.

TABLE II
AVERAGE ACCURACY AT DIFFERENT BUFFER RADII

Buffer radius (m)	Precision	Recall	F1 score
0.5	28.28%	28.07%	28.18%
1	57.00%	56.58%	56.79%
1.5	70.56%	70.03%	70.29%
2	79.21%	78.62%	78.92%
2.5	84.25%	88.66%	86.40%
3	82.74%	82.12%	82.43%
3.5	77.97%	77.39%	77.68%

III. EXPERIMENTS AND ACCURACY EVALUATION

A. Results and Comparison Between EUMAP and SOTA

We applied EUMAP to the whole study area, obtained the canopy data and the number of single-tree points of all *Eucalyptus*, and compared them with the results of manual interpretation in six accuracy evaluation regions to evaluate the detection accuracy. For the SOTA method, we selected a sample area of the same size as the accuracy evaluation region in the test area, with a total of 2000 sample points, which is similar to other deep-learning applications [17], [41]. The result extracted by SOTA is a small patch independent of each other, representing the point of the tree. In Fig. 10, the first column shows the satellite image, the second and third columns show the results of EUMAP, and the fourth column shows the SOTA mapping results.

By comparing EUMAP results from the EUMAP in Regions 1–6 with reference canopy by visual interpretation, the canopy mapping accuracy indices were calculated (see Table III). Average precision and recall were 86.47% and 90.42%, respectively. Canopy mapping accuracy for *Eucalyptus* was generally better, with $F1$ score within 85.27%–90.09%. Mapping errors resulted mainly from two factors. First, bushes and grasses in the study area cause an overmapping of the canopy. This overmapping issue can distort the accuracy of the overall mapping results, particularly in areas with much vegetation understory. Second, the shape and boundaries of small canopies are more difficult to map accurately, resulting in an underestimation or even omission of their presence.

By comparing EUMAP and SOTA results for single-tree mapping with single tree by visual interpretation in Regions 1–6, the single-tree mapping accuracy indices were calculated (see Table IV). The EUMAP average precision and recall were 84.25% and 88.66%, respectively. Also, single-tree mapping average $F1$ score was 86.40%. Analysis of the mapping results showed that the presence of additional tree species and clearings had a lower impact on single-tree mapping than on canopy mapping, whereas bushes and grasses also caused an increase in incorrect single-tree mapping and a decrease in precision. Moreover, trees with low or sparse canopies were more likely missed by the mapping algorithm, thereby inducing a decrease in recall values. Compared with the SOTA, benefit by erosion and iteration separates high canopy density forests, EUMAP improved the average $F1$ score by 12.88%. In addition to the improvement of accuracy, another advantage of EUMAP compared with deep learning is the efficiency of data processing.

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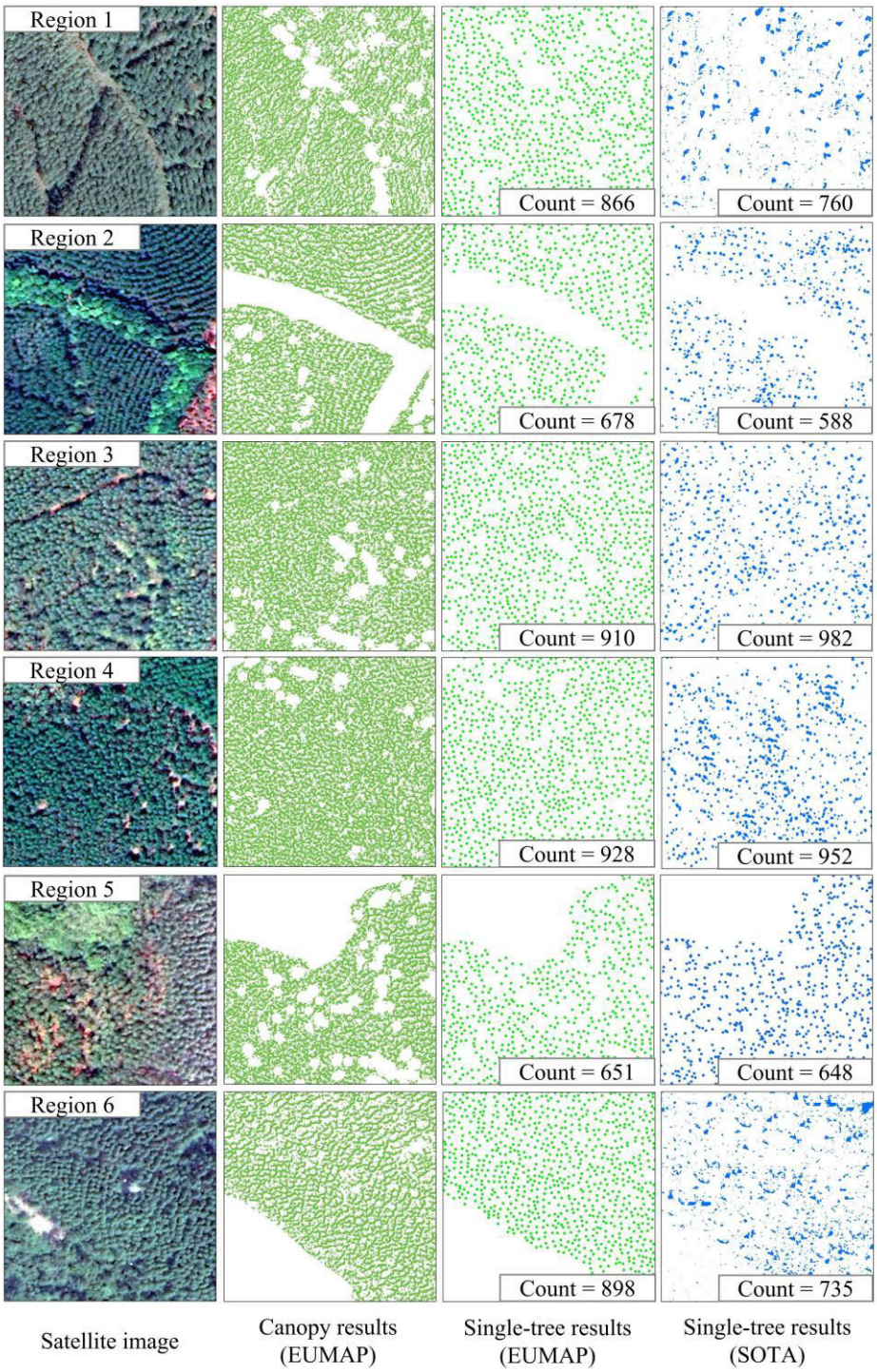


Fig. 10. Results for canopy and single-tree mapping in the accuracy evaluation regions, showing the (second column) EUMAP canopy mapping, (third column) EUMAP single-tree mapping, and (fourth column) SOTA results.

In terms of efficiency, we listed the processing process and corresponding time cost of EUMAP and SOTA in the test area under the same processor performance (see Table V). For the same image, SOTA first needs a long sample training process, which may take days or even weeks, depending on the size and complexity of the training samples. For our experiment, EUMAP does not need to spend time on training samples. In addition, SOTA recognizes a 41.4-km^2 area at a speed of about 7 h based on the trained model, which is much slower than EUMAP's overall processing speed of 26 min.

B. Region Extension Mapping

We applied EUMAP to Chengzhong and Liubei, Liuzhou, for large-scale *Eucalyptus* mapping. Based on our study result, there were a total of 7 033 021 *Eucalyptus* across the

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TABLE III
CANOPY MAPPING ACCURACY RESULTS IN THE SIX ACCURACY EVALUATION REGIONS

Region	TP (m^2)	FP (m^2)	FN (m^2)	Precision	Recall	F1 score
1	6456	356	1065	85.84%	94.77%	90.09%
2	4407	452	1071	80.45%	90.70%	85.27%
3	6503	798	723	85.94%	91.68%	88.72%
4	6918	628	1132	89.99%	89.07%	89.53%
5	3828	652	548	87.48%	85.45%	86.45%
6	5107	515	624	89.11%	90.84%	89.97%
Average				86.47%	90.42%	88.34%

TABLE IV
SINGLE-TREE MAPPING ACCURACY RESULTS IN THE SIX ACCURACY EVALUATION REGIONS

Region	EUMAP Precision	EUMAP Recall	EUMAP F1 score	SOTA Precision	SOTA Recall	SOTA F1 score
1	93.08%	91.69%	92.38%	72.24%	64.36%	68.07%
2	91.09%	90.83%	90.96%	87.59%	70.07%	77.85%
3	82.10%	90.22%	85.97%	79.71%	73.91%	76.70%
4	78.56%	87.28%	82.70%	72.30%	71.00%	71.64%
5	76.50%	78.49%	77.48%	72.99%	70.81%	71.88%
6	85.24%	91.31%	88.17%	79.48%	70.97%	74.98%
Average	84.25%	88.66%	86.40%	77.38%	70.19%	73.52%

TABLE V
EUMAP AND SOTA PROCESSING PROCESS AND CORRESPONDING TIME CONSUMING

	Same conditions	Process	Time	Total time
EUMAP	CPU: I7-10700 (8 cores, 16 threads, 2.90 GHz) Memory: 16GB (2666 MHz) Test area size = 41.4 km^2 Image resolution ≈ 0.3 m Date capacity = 6.24 GB	Canopy identification in different canopy density areas.	8 minutes	26 minutes
		Adaptive iterative erosion to separate single-tree	18 minutes	
SOTA		Sample point training	47 hours	54 hours
		Target recognition based on the trained model	7 hours	

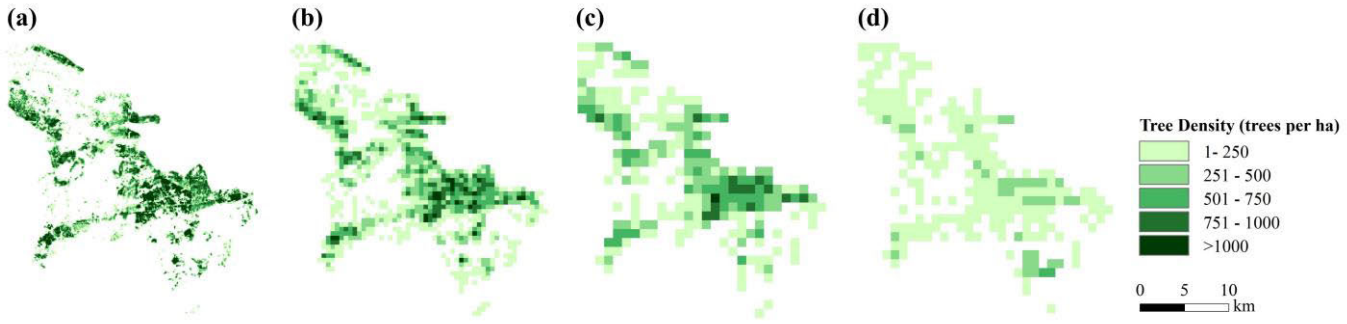


Fig. 11. *Eucalyptus* density distribution in the study area. (a)–(c) EUMAP results with spatial resolutions are 100, 500, and 1000 m, respectively. (d) Result with a spatial resolution of 1000 m, taken from [13].

study area at an average density of 819 trees per hectare. Furthermore, we mapped a series of tree density distribution maps, with the spatial resolution of 100 m, 500 m, and 1 km [see Fig. 11(a)–(c)]. Comparing tree density maps with different resolutions, it can be clearly found that as the spatial resolution increased, the spatial distribution of

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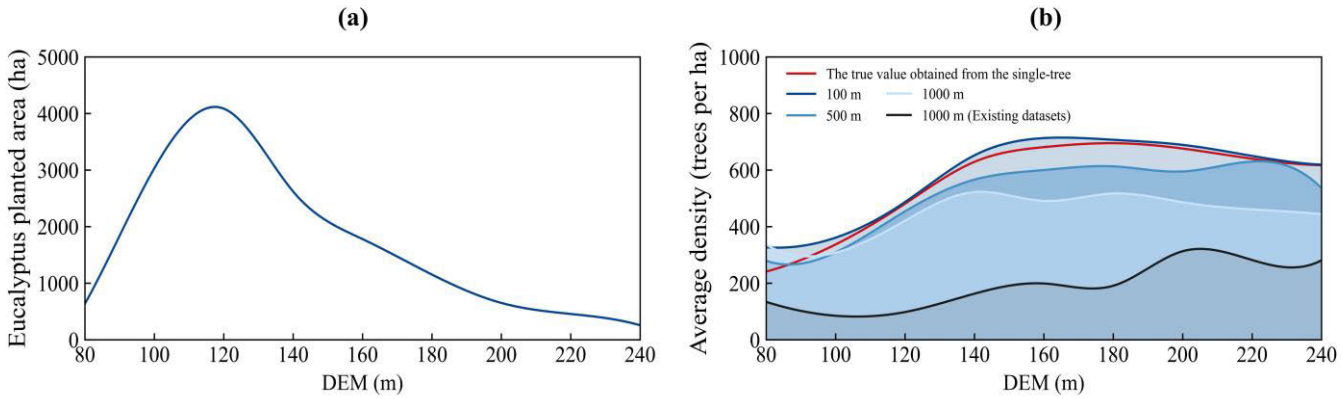


Fig. 12. Distribution and density of *Eucalyptus* on different DEMs. (a) Planted area of *Eucalyptus* on different DEM. (b) Average density of *Eucalyptus* with DEM at different resolutions. The true value is the *Eucalyptus* density calculated from the single-tree data. The existing datasets are tree density data with 1000-m resolution obtained from [13].

Eucalyptus density changed slightly. Compared to a global tree density map with a spatial resolution of 1 km in existing data products [13] [see Fig. 11(d)], the tree density map at the same resolution drawn by EUMAP provided a more detailed density distribution and a larger tree density value, which is more suitable for further precise analysis for *Eucalyptus* plantations.

In addition, we compared the distribution of differences between these four *Eucalyptus* density results of various resolutions with digital elevation model (DEM). NASA Shuttle Radar Topography Mission (SRTM) DEM product with 30-m resolution [59] was used to visualize the distribution of *Eucalyptus* planting area and the average density of *Eucalyptus* changes in various elevations. The result showed that, during the interval of 80–120 m, the planting areas increased with the increase of elevation. Then, the planting areas gradually decreased when the elevation exceeded 120 m [see Fig. 12(a)]. The *Eucalyptus* plantations in the study area are mainly distributed between 100 and 180 m, accounting for 86% of the total planting areas. Moreover, the average tree density increased with the increase in elevation, and the change tended to be flat when the elevation exceeded 140 m [see Fig. 12(b)].

The resolution of *Eucalyptus* density has a great influence on the average density of *Eucalyptus*. When the resolution of *Eucalyptus* density is 100 m, the average tree density is higher than the average tree density at other resolutions and closest to the *Eucalyptus* tree density calculated from the single-tree data. As the resolution decreases, the tree density map gradually cannot satisfy the fine comparison with the 30-m resolution DEM. The incompatibility of resolution will directly lead to the accurate decrease of the average *Eucalyptus* density. The tree density of existing data products only has a resolution of 1000 m, much smaller than the average tree density obtained by our study [13].

IV. Discussion

A. Comparison With Previous Studies

Misclassification and omission are serious sources of error for traditional extraction methods. Deep-learning methods are more accurate but require larger study datasets of higher quality. Compared with traditional mapping and deep-learning methods, this study proposes an approach to canopy and single-tree mapping (EUMAP), relying on image morphology and physical features of *Eucalyptus*. EUMAP accounts for texture and spectral features, by extracting and processing palmprint-like or fingerprint-like imagery information of *Eucalyptus*, ultimately achieving satisfactory results. On the one hand, applying Otsu's binarization method to image regions with distinct canopy densities, separated by morphological opening operations, minimizes the negative impact of spectral differences between these regions on the deep-learning identification process [60]. This allows for effective canopy and single-tree mapping in both dense- and sparse-canopy areas. On the other hand, *Eucalyptus* single-tree canopy dimensions are not uniform; frequent connections and overlaps between single-tree canopies complicate the mapping process [61]. In this study, the adaptive iterative erosion algorithm simultaneously solved the problem of canopy density heterogeneity and the problem of connecting or overlapping canopies. Compared to the SOTA method, EUMAP demonstrates superior performance in terms of faster efficiency and higher accuracy. The single-tree points generated by EUMAP provide a more detailed and accurate description for the distribution and density of *Eucalyptus*. Also, the *Eucalyptus* density map generated by EUMAP exhibits higher quality and practicality compared to existing data sets.

B. Importance of This Study and Remaining Sources of Uncertainty

This study shows that the proposed method is directly applicable to high-resolution satellite images, implying simple access to the data source and no necessity of using drones for image acquisition [62], [63]. Moreover, the spectral domain and image resolution requirements are easily fulfilled because this approach is directly applicable to the conventional RGB bands of optical images. This adaptability lends itself to a broader range of applications. Based on existing morphological processing knowledge and proven methods, EUMAP is able to accurately map single tree by extracting and processing textural and spectral features of *Eucalyptus* plantation areas and single *Eucalyptus*. Most importantly, EUMAP does not need to conduct sample selection, model training, and image

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classification, which could significantly improve the mapping efficiencies. In addition, EUMAP not only clearly explains the process of image feature processing in single-tree mapping but also can rapidly map millions of single trees over the large-scale region. Therefore, the scalability and practicality of the method are enhanced. The study extends to the domain of ecological research because it provided important information on the ecological structure and physical parameters of *Eucalyptus* plantations. This information is essential to balance the ecological and economic aspects of *Eucalyptus* plantation exploitation to achieve sustainability [64], [65]. The number of trees in representative areas could also constitute a meaningful indicator for forest management practices and decision-makers in the governmental and private sectors [66], [67]. In conclusion, the results in this study add scientific and technical values to the aspects covering the applicability of satellite imagery, efficiency of data processing, real-time change monitoring, and forest management. Although this study clearly improves canopy and single-tree mapping, there are still uncertainties in the analysis methodology. First, additional field work is needed for the analysis to include not only the number of *Eucalyptus*single tree but also the planting year and tree structure. Second, although cloud cover and light intensity also affect remote sensing image quality, our test data directly included WorldView-3 satellite images, without image preprocessing such as radiation correction or cloud removal. In addition, the proposed algorithm parameters were determined by rigorous testing, but variations of *Eucalyptus* growth between satellite images could introduce errors in parameter selection for canopy and single-tree mapping. During erosion processes, smaller than 0.5-m^2 canopy areas are also neglected, and they have few recognizable features on 0.3-m resolution images. To address these issues, further study should focus on error mitigation in terms of data quality and automatic parameter optimization. In addition, the prior knowledge of the spatial distribution of *Eucalyptus* plantation is the key to successfully implement our proposed method to accurately count the numbers of *Eucalyptus.* Further study is needed to combine the *Eucalyptus* plantations identification algorithm and single-tree mapping approach since it can help us address deeper scientific question on counting the numbers of *Eucalyptus* and monitoring their spatial and temporal changes [68].

V. Conclusion

In this study, an adaptive iterative erosion approach (EUMAP) was proposed to analyze submeter-resolution satellite images for *Eucalyptus* forest canopy and single-tree mapping. EUMAP provides canopy and single-tree information by separating image regions with different canopy densities and adapting the number of iterations of the erosion process to each region. Compared with field survey data from the State-owned Sanmenjiang forest farm in Liuzhou, China, EUMAP achieves an average $F1$ score of 88.34% in canopy mapping and 86.40% in single-tree mapping. In addition, we manually located more than 7 000 000 *Eucalyptus* single tree in study area from easily accessible and free remote sensing images without training samples. Our study improves the mapping accuracy and data processing efficiency for mapping *Eucalyptus*plantation's canopy and single-tree information. It has been confirmed that EUMAP can be used for regional promotion in wider area, which provides strong support for large-scale and low-cost forest mapping. Moreover, the canopy and single-tree information obtained by EUMAP can be used for estimating forest cover and forest carbon cycle and helping to achieve fine management, protection, and utilization of forest resources. In the future, with the support of multitemporal or multisource remote sensing data, long time series monitoring of trees might greatly assist us in accurately analyzing the forest dynamic, carbon stock, and natural disasters.

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